

FPGAs IN DATA CENTRES: Opportunities And Challenges

Data centres play a major role in expanding the enterprise computing capabilities. These are centralised repository for storage, management and dissemination of data and information. Today's central processing units (CPUs) are powerful and well-suited for high-level decision-making and control tasks, but their computing power cannot be scaled economically to keep pace with ever-increasing data centre requirements.

Hardware acceleration is a solution which may be implemented using a variety of technologies, including application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), graphical processing units (GPUs) and network processing units (NPUs). FPGAs are well known for their versatility. Many of these implement communication and control functions wholly, or in part, as soft intellectual property (IP). So in data centres FPGAs can play a role in cognitive computing, high-performance computing and the Internet of Things. These can serve as coprocessors to accelerate CPU workloads. This is an approach used in supercomputing and high-performance computing (HPC), done usually by teaming CPUs with Nvidia GPUs or Intel's x86-based Xeon Phi coprocessors.

Cloud service systems are based upon processors. Software is driving the change, which extends beyond the rise of machine learning and training neural models. A whole new set of workloads is running in the cloud today, including virtualising net-

works, storage processing, compression and encryption. FPGAs slot into the standard server, drawing 25W from its PCI Express (PCIe) Gen 3 slot.

An integrated platform is one in which FPGA developers can upload their design in an IP library and data centre users can download the IP cores they need. This decoupling will enable widespread adoption of heterogeneous processing in data centres. Data centre operators can optimise the infrastructures by hyperscale, where FPGAs offer more sophisticated processing architectures that bring HPC-style parallel processing of workloads.

Accelerator basics

Accelerating the data centre applications has become a requirement due to power and space constraints. Accelerators are needed to increase performance at a low total cost of ownership for targeted workloads.

An accelerator is a separate architectural substructure on the same chip or on a different die. It is architected with a different set of objectives than the base processor, where objectives are based on the needs of a special class of applications.

Accelerators are tuned to provide a higher performance at lower cost, or at lower power, or with less development effort than with the general-purpose base hardware. Depending on the domain, accelerators often bring greater than ten times advantage in performance, cost or power consumption over a general-purpose processor.

Examples of accelerators include



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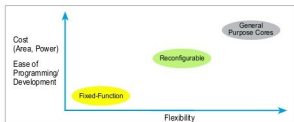


Fig. 1: Accelerator architecture

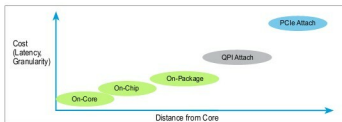


Fig. 2: Accelerator attach